2023春高数下期中试题(回忆版)

1. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) If P_1 and P_2 are the following planes:

$$P_1 := \{(x,y,z) : x+2y-z = \frac{\sqrt{2}}{2}\},$$
$$P_2 := \{(x,y,z) : x+y = -\frac{\sqrt{2}}{2}\}.$$

then the angle θ between P_1 and P_2 is:

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$. (D) $\frac{\pi}{6}$.

(2) The arc length of the curve $r(t) = \frac{2}{3}t^{\frac{3}{2}}i + \frac{2}{3}(2-t)^{\frac{3}{2}}j + (t-1)k$ for $\frac{1}{4} \le t \le \frac{1}{2}$ is

(A) $\frac{1}{4}$ (B) $\frac{\sqrt{2}}{4}$ (C) $\frac{\sqrt{3}}{4}$ (D) $\frac{1}{2}$.

(3) Let $f(x,y) = \begin{cases} \frac{\ln(1+xy)}{x}, & x \neq 0 \\ a, & x = 0 \end{cases}$, then

(A) If $a=1$, then $f(x,y)$ is continuous at $(0,2)$.

(B) If $a=2$, then $f(x,y)$ is discontinuous at $(0,2)$.

(C) If $a=1$, then $f(x,y)$ is continuous at $(0,1)$.

(D) If $a=2$, then $f(x,y)$ is continuous at $(0,1)$.

(4) If $0 \le a_n \le \frac{1}{n}$ $(n=1,2,\cdots)$, which of the following series must be convergent?

(A) $\sum_{n=1}^{\infty}(-1)^n \frac{a_n}{(1+\ln n)^2}$ (B) $\sum_{n=1}^{\infty}(-1)^n a_n$

(C) $\sum_{n=1}^{\infty} a_n$. (D) $\sum_{n=1}^{\infty}(-1)^n \frac{a_n}{1+\ln n}$

(5) If $\sum_{n=1}^{\infty} a_n x^n$ converges at $x=-2$, then

(A) $\sum_{n=1}^{\infty} a_n 2^n$ converges. (B) $\sum_{n=1}^{\infty} a_n 2^n$ diverges.

(C) $\sum_{n=1}^{\infty} n a_n$ converges (D) $\sum_{n=1}^{\infty} n a_n$ diverges.

2. Fill in the blanks.

(1) Let the polar equation be $r = \csc\theta e^{r\cos\theta}$. Its equivalent Cartesian equations is _____.

(2) The distance from the point $S(1,1,1)$ to the plane $x+2y+3z+4=0$ is _____.

(3) A particle is traveling with acceleration $a(t) = \langle e^t, t, \sin 2t\rangle$. At $t=0$, it was at the origin and its initial velocity is $v(0) = \langle 1,0,0\rangle$. The position function of the particle is ____.

(4) $\lim_{x\to 0} \frac{3\sin(2x) - 2\sin(3x)}{6x^3 + x^4} =$ _____.

(5) The series $$S = -\ln 2 + \frac{\ln^2 2}{2!} + \cdots + \frac{(-1)^n \ln^n 2}{n!} + \cdots$$

converges and has the sum _____.

3. (1) Find the interval of convergence for the series $\sum_{n=1}^{\infty} \frac{\ln n}{n}(x-1)^n$.

(2) For what value of x the above series converges absolutely, or conditionally?

4. Find the limit if it exists. If it does not exist, explain why.

(1) $\lim_{(x,y)\to(0,0)} \frac{y^2\sin^2 x}{x^4+y^4}$.

(2) $\lim_{(x,y)\to(0,0)} \frac{2xy}{\sqrt{x^2+2y^2}}$

5. Let $L_1: x=t, y=1, z=1+t$ and $L_2: x=1+2s, y=0, z=2s$. Is there one plane in which both lines lie? If so, find the equation of the plane. If not, give your reason.

6. Assume that a and b are real numbers, and $0<b<a<1$.

(a) Determine whether the series.

$$\frac{a}{\sqrt{(1+a)(1+b)}} + \frac{b^2}{\sqrt{(1+a^2)(1+b^2)}} + \frac{a^3}{\sqrt{(1+a^3)(1+b^3)}} + \frac{b^4}{\sqrt{(1+a^4)(1+b^4)}} + \cdots + \frac{c_n}{\sqrt{(1+a^n)(1+b^n)}} + \cdots,$$

where $c_n = \begin{cases} a^n, & \text{if } n \text{ is odd.} \\ b^n, & \text{if } n \text{ is even.} \end{cases}$

converges or diverges. Justify your answer!

(b) Find the limit: $\lim_{n\to\infty}\left(\frac{a^n}{n^2}+\frac{b^n}{n}\right)^{\frac{1}{n}}$

7. Let C be the curve given by

$$r(t) = \sqrt{2}\cos t\, i + \sin t\, j + \sin t\, k.$$

(a) Find the unit tangent vector $T(t)$ and principal unit normal vector $N(t)$ for C.

(b) Find the curvature $k(t)$.

8. Using known Taylor series expansions, write the Taylor series of for the function.

$$f(x) = \frac{4}{x^2-2x+5} + \ln x$$

at $x=1$ in the interval $(0,2)$.